

Business Process Document

Rwanda NBSAP Monitoring Dashboard 2025–2030

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Document Control

Field	Value
Document Title	Business Process Document
Project	Rwanda NBSAP Monitoring Dashboard
Prepared by	NBSAP Dashboard Development Team
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1. Introduction

1.1 Purpose

This document describes the business processes automated by the Rwanda NBSAP Monitoring Dashboard. It maps the end-to-end workflows from data collection through verification to national reporting, identifying process owners, decision points, and system interactions at each stage.

1.2 Process Landscape

The NBSAP Monitoring Dashboard supports the following core business processes:

Process ID	Process Name	Process Owner
BP-01	Data Collection & Submission	Ministry/District Reporters
BP-02	Data Verification & Approval	REMA Administrators / Lead Ministry Reviewers
BP-03	Automated Metric Calculation	System (database triggers)

BP-04	National Target Progress Monitoring	REMA Senior Management
BP-05	User Account & Role Management	REMA Administrators
BP-06	Role Change Request Workflow	System Users / REMA Administrators
BP-07	Data Export & CBD Reporting	REMA Reporting Officers
BP-08	Geographic & Biodiversity Data Integration	System (automated APIs)
BP-09	Risk & Compliance Management	REMA / Policy Monitors

2. Business Process Descriptions

BP-01: Data Collection & Submission

Process Owner: Ministry Reporters / District Reporting Officers

Objective: Collect structured biodiversity data from field operations, institutional activities, and sectoral programmes through standardised reporting tools.

Trigger: Reporting period deadline or occurrence of reportable event (e.g., HWC incident)

Process Flow:

```

START
|
|— Reporter logs into the system
|
|— Reporter selects appropriate Reporting Tool (T01–T07)
|   |— T01: National Institutional Reporting
|   |— T02: Ecosystem & Habitat Monitoring
|   |— T03: Protected Area Management
|   |— T04: Human-Wildlife Conflict
|   |— T05: Finance & Resource Mobilisation
|   |— T06: Private Sector EIA Compliance
|   |— T07: Community Engagement
|
|— Reporter completes structured form fields
|   |— Selects district (auto-populated for district officers)
|   |— Selects NBSAP target alignment
|   |— Enters quantitative data (hectares, counts, budgets)
|   |— Enters qualitative observations
|
|— Reporter attaches supporting evidence (documents, images)
|
|— System validates form data (required fields, value ranges)
|   |— PASS → Submission created with status 'pending'
|   |— FAIL → Validation errors displayed; reporter corrects and resubmits

```

```
|
|
| System generates audit log entry
|
| System sends notification to authorised reviewers
|
END (Report enters Verification Queue)
```

Business Rules:

- Only users with `canSubmitReports` permission may submit reports
- District Reporting Officers are restricted to their own district
- All form fields have defined validation rules (type, range, required)
- Evidence attachments are stored in secure cloud storage (Supabase Storage)
- Submissions default to `pending` status — no data affects national metrics at this stage

BP-02: Data Verification & Approval

Process Owner: REMA Administrators / Lead Government Ministry Reviewers

Objective: Ensure all biodiversity data meets quality standards before it updates national metrics. This is the single most critical quality gate in the system.

Trigger: New submission appears in the Verification Queue

Process Flow:

```
START
|
| Reviewer opens the Verification Queue
|
| Reviewer selects a pending submission
|
| Reviewer examines:
|   | Form data (completeness, plausibility, consistency)
|   | Attached evidence (documents, images)
|   | Historical context (previous submissions for same district/tool)
|   | Reporter credentials (organisation, role)
|
| DECISION POINT: Is the data acceptable?
|   |
|   | YES → Reviewer clicks "Approve"
|   |   | Status changes to 'approved'
|   |   | Database triggers fire → national metrics auto-update (BP-03)
|   |   | Notification sent to reporter: "Your report was approved"
|   |   | Audit log entry created
|   |
|   | NO → Reviewer clicks "Reject"
|   |   | Reviewer enters rejection reason (mandatory)
```

```
|
|   └─ Status changes to 'rejected'
|   └─ Notification sent to reporter with rejection reason
|   └─ Reporter may revise and resubmit (returns to BP-01)
|   └─ Audit log entry created
|
END
```

Business Rules:

- Only users with `canApproveReports` permission may verify reports
- Rejection requires a written reason (mandatory field)
- Approved data immediately and automatically updates system metrics
- The reviewer's identity and timestamp are recorded on the report
- A reviewer cannot approve their own submission (separation of duties)

BP-03: Automated Metric Calculation

Process Owner: System (fully automated — no human intervention)

Objective: Maintain accurate, real-time national biodiversity metrics by automatically computing aggregated values from approved report data.

Trigger: A report is approved (BP-02) OR a previously approved report is deleted

Process Flow:

```
START
|
| └─ Database trigger fires on report status change to 'approved'
|
| └─ System identifies the report's tool type (T01-T07)
|
| └─ System extracts relevant metric values from form_data JSONB
|   └─ Forest hectares restored (T02)
|   └─ Wetland hectares restored (T02)
|   └─ Protected area coverage (T03)
|   └─ HWC incidents recorded (T04)
|   └─ Budget disbursed (T05)
|   └─ EIA compliance rate (T06)
|   └─ Community participants (T07)
|
| └─ System updates the system_metrics table
|   └─ Increment/decrement metric values
|   └─ Recalculate percentages and averages
|   └─ Update last_updated timestamp
|
| └─ System recalculates target progress using tool weights
|   └─ Each tool (T01-T07) has a configurable weight
```

```
| | | Weighted progress = SUM(tool_progress × tool_weight)
| | | Target progress bar updated
| |
| | Websocket pushes real-time update to all connected browsers
|
END
```

On Report Deletion (Reversal):

```
| Database trigger fires on deletion of an approved report
| System reverses all metric contributions of the deleted report
| Recalculates affected target progress
| Audit log entry created: "Metric reversal – Report [ID] deleted"
```

Business Rules:

- No manual metric calculation is permitted; all values are trigger-driven
- Each reporting tool has a configurable weight (stored in `tool_weights` table)
- 13 system-wide metric types are maintained automatically
- Metric reversal on deletion ensures data consistency

BP-04: National Target Progress Monitoring

Process Owner: REMA Senior Management / Policy Monitors

Objective: Provide executive-level visibility into Rwanda's progress against all 22 NBSAP targets.

Process Flow:

```
START
|
| Authorised user accesses the Dashboard
|
| System displays:
| | | 22 national targets organised by 4 KM-GBF Goals
| | | Progress bars with 3-colour status (green/amber/red)
| | | 13 automated system metrics
| | | Recent approved reports
| | | District-level compliance map
|
| User drills down into specific targets
| | | Linked indicators with baseline and current values
| | | Implementation milestones (2025–2030 timeline)
| | | Responsible stakeholders
| | | Contributing report data
|
| User may export data for external use (BP-07)
```

END

Progress Colour Coding:

Colour	Threshold	Meaning
Green	>= 70%	On track to meet 2030 target
Amber	40% – 69%	At risk — intervention may be needed
Red	< 40%	Behind schedule — urgent attention required

BP-05: User Account & Role Management

Process Owner: REMA Administrators

Objective: Manage the lifecycle of user accounts, ensuring appropriate access levels and institutional accountability.

Process Flow:

```
START
|
|— New user registers via the authentication portal
|   |— Default role: Policy Monitor (read-only)
|   |— Account status: Active
|
|— REMA Administrator reviews new account
|   |— Assigns appropriate role based on institutional mandate
|   |— Updates organisation and department fields
|   |— Audit log entry created
|
|— ONGOING MANAGEMENT:
|   |— Suspend account (with reason and optional end date)
|       |— Auto-reactivation on end date if set
|   |— Deactivate account (permanent)
|   |— Change user role
|       |— Review login activity
|
END
```

BP-06: Role Change Request Workflow

Process Owner: System Users (requesters) / REMA Administrators (approvers)

Objective: Provide a formal, auditable process for users to request elevated or changed access levels.

Process Flow:

```
START
|
|— User submits role change request
|   |— Selects desired role from available options
|   |— Provides justification text
|   |— Request created with status 'pending'
|
|— REMA Administrator reviews request
|   |— Evaluates justification and institutional need
|   |
|   |— APPROVE → User's role is updated immediately
|   |   |— Notification sent to user
|   |   |— Audit log entry created
|   |
|   |— REJECT → Request marked as rejected
|   |   |— Rejection reason recorded
|   |   |— Notification sent to user
|   |   |— Audit log entry created
|
|
END
```

BP-07: Data Export & CBD Reporting

Process Owner: REMA Reporting Officers

Objective: Generate structured data exports suitable for internal analysis, government reporting, and international CBD obligations.

Process Flow:

```
START
|
|— REMA officer navigates to Reports page
|
|— Selects export scope:
|   |— By reporting tool (T01-T07)
|   |— By date range
|   |— By district
|   |— By NBSAP target
|   |— Full dataset
|
|— Selects export format:
|   |— CSV – Raw data for spreadsheet analysis
```

```
|
|   └─ PDF – Formatted report for printing/distribution
|   └─ JSON – Structured data for system interoperability
|
└─ System generates export file
|
└─ User downloads file
|
└─ For CBD Reporting:
|   └─ Indicator data maps to KM-GBF 3-tier structure
|   └─ Headline indicators → Tier 1 (internationally comparable)
|   └─ Component indicators → Tier 2 (implementation detail)
|   └─ Binary indicators → Tier 3 (policy condition assessments)
|
END
```

BP-08: Geographic & Biodiversity Data Integration

Process Owner: System (automated)

Objective: Provide real-time geographic and species data from external sources to enrich dashboard visualisations.

Process Flow:

```
START
|
└─ GBIF Integration (species occurrence data):
|   └─ System queries GBIF API (country=RW)
|   └─ Rate-limited to 1 request/second
|   └─ Proxied via Supabase Edge Function (CORS avoidance)
|   └─ Results displayed on interactive map
|   └─ Auto-refresh every 30-60 seconds
|
└─ RBIS Integration (Rwanda Biodiversity Information System):
|   └─ 8 predefined data streams monitored
|   └─ Connection status tracked (connected/disconnected/error)
|   └─ Linkage to specific NBSAP indicators maintained
|   └─ Connection events logged for audit
|
└─ GeoJSON Layers (static reference data):
|   └─ Rwanda district boundaries (30 districts)
|   └─ Protected area polygons
|   └─ River network lines
|   └─ Lake boundaries
|
END
```

BP-09: Risk & Compliance Management

Process Owner: REMA / Policy Monitors

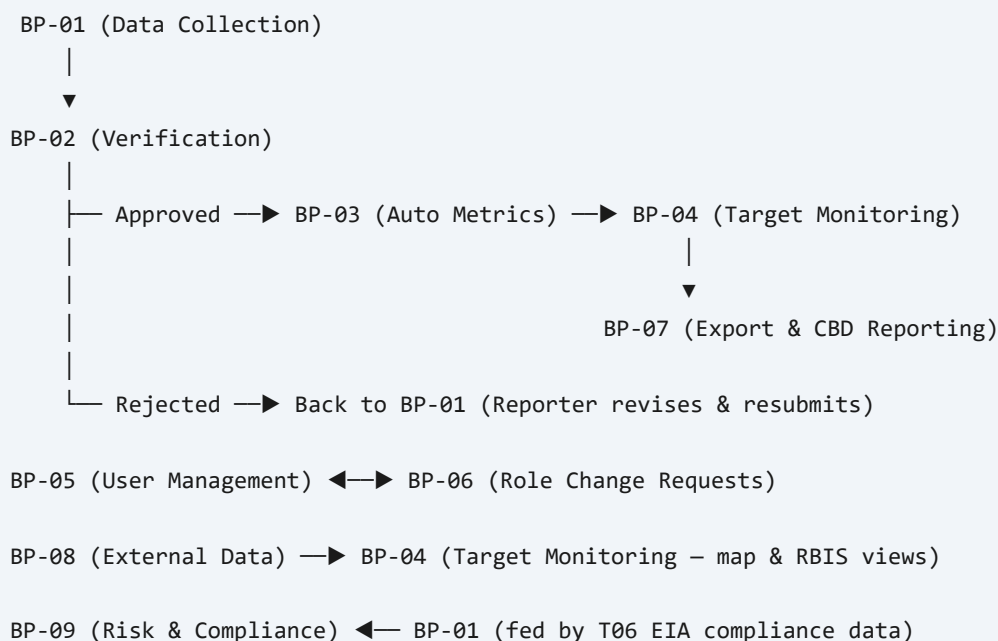
Objective: Identify, track, and mitigate risks to biodiversity targets, and monitor compliance with national and international frameworks.

Process Flow:

```
START
|
|─ Risk Register:
|   |─ Risks categorised by type and severity (High/Medium/Low)
|   |─ Each risk has: description, likelihood, impact, mitigation, owner
|   |─ Active risks flagged for dashboard visibility
|   |─ Risk status reviewed periodically
|
|─ Compliance Tracking:
|   |─ EIA compliance monitored per district and sector
|   |─ Compliance scores auto-calculated from T06 submissions
|   |─ District-level compliance visible on map
|   |─ Non-compliance flagged for follow-up
|
END
```

3. Process Integration Map

The following diagram shows how the 9 business processes interact:



4. Key Performance Indicators (KPIs)

KPI	Target	Measurement Source
Average report verification turnaround	< 48 hours	Verification Queue timestamps
Percentage of districts with 'submitted' status	> 90%	District compliance tracking
Number of reports submitted per quarter	> 50 across all tools	Toolkit report counts
System uptime	99.5%	Vercel/Supabase monitoring
Audit log completeness	100% of system actions logged	Audit log table
CBD export readiness	Indicator data exportable in 3-tier structure	Export functionality

5. Process Improvement Opportunities

Opportunity	Current State	Future State	Priority
Automated notifications for overdue reports	Manual follow-up	Scheduled reminders to districts with 'missing' status	High
Multi-language support	English only	Kinyarwanda and French translations	Medium
Mobile application	Responsive web only	Native iOS/Android app for field data collection	Medium
Offline data collection	Online only	Offline-capable forms with sync-when-connected	Medium
Automated CBD report generation	Manual compilation from exports	One-click CBD National Report draft	High

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